Day 66 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in Science & Technology (3): Quantum Computing & Semiconductors

READING MATERIAL

Covers India's recent quantum computing and semiconductor milestones — a first full-stack quantum computer, the country's first quantum technology park, and progress toward domestic chip production.

Q: What is QpiAI-Indus, and when was it launched?

A: India's first full-stack quantum computing system — a 25-qubit quantum computer combining advanced quantum hardware, scalable control, and optimised software. Launched in April 2025 by the Indian startup QpiAI.

Q: What is the Quantum Valley Tech Park, where is it located, and when was it established/set to open?

A: India's FIRST quantum technology park, established in May 2025 in Amaravati, Andhra Pradesh — scheduled to open in January 2026. IBM, TCS, and the Government of Andhra Pradesh have partnered to develop India's largest quantum computer there.

Q: What are the National Quantum Mission's key technical objectives?

A: To develop intermediate-scale quantum computers with 50-1000 physical qubits within 8 years (across platforms like superconducting and photonic technology), plus satellite-based secure quantum communication between ground stations over a range of 2000 km within India, and longer-distance international quantum communication links.

Q: Which two institutions host key quantum infrastructure facilities, and what does each focus on?

A: IIT Bombay hosts a Quantum Sensing & Metrology facility. IISc Bengaluru hosts a Quantum Computing fabrication facility, supporting manufacturing of quantum computing chips (superconducting, photonic, and spin qubit-based).

Q: What is the India Semiconductor Mission 2.0, and what milestone was anticipated by the end of 2025?

A: A broadened mission supporting the entire semiconductor value chain, expected to receive Cabinet approval by October 2025. The first domestically produced semiconductor chip was anticipated by the end of 2025.

MOCK TEST (WITH ANSWERS)

1. QpiAI-Indus, India's first full-stack quantum computer, has how many qubits?

A. 10

B. 25 ✓

C. 50

D. 100

Explanation: 25 qubits. [medium, web-research]

2. QpiAI-Indus was launched by the startup QpiAI in which month/year?

A. January 2025

B. April 2025 ✓

C. August 2025

D. January 2026

Explanation: April 2025. [medium, web-research]

3. India's first quantum technology park, the Quantum Valley Tech Park, is located in:

A. Bengaluru, Karnataka

B. Hyderabad, Telangana

C. Amaravati, Andhra Pradesh ✓

D. Pune, Maharashtra

Explanation: Amaravati, Andhra Pradesh. [medium, web-research]

4. The National Quantum Mission aims to develop quantum computers with how many physical qubits within 8 years?

A. 10-50

B. 50-1000 ✓

C. 1000-5000

D. Over 10,000

Explanation: 50-1000 qubits. [hard, web-research]

5. The National Quantum Mission's satellite-based secure quantum communication aims to cover ground stations over a range of:

A. 500 km

B. 1000 km

C. 2000 km ✓

D. 5000 km

Explanation: 2000 km within India. [hard, web-research]

6. Which institution hosts a Quantum Computing fabrication facility supporting chip manufacturing?

A. IIT Bombay

B. IISc Bengaluru ✓

C. IIT Delhi

D. IIT Madras

Explanation: IISc Bengaluru. [medium, web-research]

Part B – Aptitude & Mental Ability: Percentage

READING MATERIAL

Percentage is 1 of 10 named Aptitude sub-skills. TNPSC's percentage questions are built from a handful of recurring shapes — successive discounts, reverse percentage ('20% of x = 50, find x'), percentage of a mixture, and percentage of a time period — recognize the shape first, then compute.

Q: Method — successive percentage discounts: A TV was sold for ₹14,400 after successive discounts of 10% and 20%. What was its original (market) price?

A: Successive discounts don't add (it's NOT a flat 30% off) — apply them one after another. If the market price is P: P × 0.90 × 0.80 = 14,400 → P × 0.72 = 14,400 → P = 14,400 / 0.72 = ₹20,000.

Q: Method — percentage of a mixture (the parts must add to 100%): An alloy is 30% copper and 40% zinc, rest nickel. Find the nickel in 20 kg of the alloy.

A: Nickel % = 100 − 30 − 40 = 30%. Nickel amount = 30% of 20 kg = 6 kg. The key step is realizing the percentages must total 100% — nickel is whatever's left over, not a number given directly.

Q: Method — reverse percentage ('find the whole, given a percentage of it'): If 20% of x = 50, find x.

A: 20% of x = 50 → 0.20 × x = 50 → x = 50 / 0.20 = 250. Reverse-percentage questions give you the part and the percentage, and ask for the whole — divide, don't multiply.

Q: Method — percentage of a decimal/small number: Find 0.75% of 40 kg.

A: 0.75% = 0.0075. 0.0075 × 40 = 0.3 kg. Common mistake: treating '0.75%' as '0.75' (i.e. 75%) instead of converting the decimal percentage properly — always convert to a decimal fraction first (divide by 100) before multiplying.

Q: Method — percentage of a time period: What percentage of a day is 10 hours?

A: A day = 24 hours. 10/24 × 100 = 41.67%.

Q: Method — percentage of a fraction: What percent of 2/7 is 1/35?

A: Set up: x% of (2/7) = 1/35 → x/100 × 2/7 = 1/35 → x = (1/35) × (7/2) × 100 = (7/70) × 100 = 10%.

Q: Method — sum of two independent percentage-of-number terms: Find the value of 28% of 450 + 45% of 280.

A: 28% of 450 = 126. 45% of 280 = 126. Sum = 126 + 126 = 252. (A nice coincidence both terms equal 126 here — don't expect that in general, just add the two percentage values separately.)

Q: Method — percentage greater than 100% (the result is bigger than the original number): Find 280% of 3,940.

A: 280% = 2.80. 2.80 × 3,940 = 11,032. Percentages over 100% just mean the result is larger than the base number — the calculation method doesn't change.

MOCK TEST (WITH ANSWERS)

1. A mobile phone was sold for ₹10,200 after successive discounts of 15% and 20%. What was its original price?

A. ₹14,500

B. ₹15,000 ✓

C. ₹15,500

D. ₹16,000

Explanation: P × 0.85 × 0.80 = 10,200 → P × 0.68 = 10,200 → P = 15,000. [medium, authored]

2. An alloy is 25% tin and 35% lead, the rest is copper. Find the copper in 40 kg of the alloy.

A. 10 kg

B. 14 kg

C. 16 kg ✓

D. 20 kg

Explanation: Copper % = 100−25−35 = 40%. 40% of 40 kg = 16 kg. [easy, authored]

3. If 35% of y = 91, find y.

A. 230

B. 250

C. 260 ✓

D. 240

Explanation: y = 91 / 0.35 = 260. [easy, authored]

4. Find 1.25% of 80 kg.

A. 1 kg ✓

B. 10 kg

C. 0.1 kg

D. 12.5 kg

Explanation: 1.25% = 0.0125. 0.0125 × 80 = 1 kg. [easy, authored]

5. What percentage of a day is 6 hours?

A. 20%

B. 25% ✓

C. 30%

D. 16.67%

Explanation: 6/24 × 100 = 25%. [easy, authored]

6. What percent of 3/8 is 3/40?

A. 10%

B. 15%

C. 20% ✓

D. 12%

Explanation: x% of (3/8) = 3/40, so x/100 = (3/40) ÷ (3/8) = (3/40) × (8/3) = 1/5, giving x = 20%. [medium, authored]

7. Find the value of 32% of 350 + 18% of 200.

A. 148 ✓

B. 138

C. 150

D. 146

Explanation: 32% of 350 = 112. 18% of 200 = 36. Sum = 148. [medium, authored]

8. Find 320% of 2,500.

A. 7,500

B. 8,000 ✓

C. 8,500

D. 9,000

Explanation: 320% = 3.2. 3.2 × 2,500 = 8,000. [easy, authored]

9. A quantity is first increased by 20%, then decreased by 20%. What is the net percentage change?

A. No change

B. 4% decrease ✓

C. 4% increase

D. 20% decrease

Explanation: Net change after +20% then −20% is always a decrease of (20×20)/100 = 4% — increasing and then decreasing by the same percentage never cancels out, because the second change applies to a different (larger) base. [hard, authored]

10. The price of an item increases from ₹250 to ₹300. What is the percentage increase?

A. 15%

B. 20% ✓

C. 25%

D. 30%

Explanation: Increase = 50. Percentage increase = (50/250) × 100 = 20%. [easy, authored]

Part C – General English: Writing Skills: Letter Writing (Formal and Informal)

READING MATERIAL

Covers the structural differences between formal letters, informal letters, and memos — exactly the 'what belongs/doesn't belong' MCQ format TNPSC uses for this topic.

Q: Real exam: Which of the following is suitable as a salutation for a FORMAL letter?

A: 'Dear Sir/Madam' — confirmed correct by a real exam question, as opposed to casual options like 'Hello', 'Cheers', or 'Best regards' (the latter is actually a CLOSING, not an opening salutation).

Q: Real exam: Which characteristic is NOT typical of a formal letter?

A: Use of slang or unprofessional expressions — a formal letter should use professional language, clear structure, and convey official information; slang is the OPPOSITE of what's expected.

Q: Real exam: Which component is typically NOT included in a memo (as opposed to a letter)?

A: A greeting/salutation — a memo typically has a Subject line, To/From sections, and a signature line, but SKIPS the formal greeting that a letter would include, since memos are brief internal communications.

Q: What are the typical parts of a FORMAL letter?

A: Sender's address (top left), date, recipient's address, a subject line summarising the purpose, a formal salutation ('Dear Sir/Madam' or 'Dear Mr./Ms. [Surname]'), the body in clear paragraphs, a complimentary close ('Yours faithfully' or 'Yours sincerely'), and a signature with name in block letters.

Q: What is typically NOT required in an INFORMAL letter, that IS required in a formal one?

A: A formal Subject line and the recipient's full official address — informal letters use a friendly greeting ('Dear [Name]' or 'Hi [Name]') and a casual closing ('Take care', 'All the best'), and the format is generally more flexible.

MOCK TEST (WITH ANSWERS)

1. Which of the following is a suitable salutation for a formal letter?

A. Hello

B. Dear Sir/Madam ✓

C. Cheers

D. Best regards

Explanation: Dear Sir/Madam. [easy, question-bank-2025]

2. Which of the following is NOT characteristic of a formal letter?

A. Use of professional language

B. Clear and understandable structure

C. Use of slang or unprofessional expressions ✓

D. Conveying official information

Explanation: Use of slang/unprofessional expressions. [medium, question-bank-2025]

3. Which of the following is NOT typically included in a memo?

A. Subject line

B. Greeting/Salutation ✓

C. To and from sections

D. Signature line

Explanation: Greeting/Salutation. [medium, question-bank-2025]

4. A formal letter typically ends with which complimentary close?

A. Take care

B. All the best

C. Yours faithfully / Yours sincerely ✓

D. Bye for now

Explanation: Yours faithfully / Yours sincerely. [easy, web-research]

5. What is typically NOT required in an informal letter, unlike a formal one?

A. A friendly greeting

B. A formal Subject line and official recipient address ✓

C. A closing remark

D. The sender's address

Explanation: A formal Subject line and official recipient address. [medium, web-research]